

## AI Applications

### Learning Objectives

- Identify AI applications across diverse industries
- Analyze how AI transforms traditional processes in specific domains
- Evaluate the societal impact of AI deployment in everyday life

### Engage (5-7 minutes)

Start with an “AI audit” of the student’s morning: phone face unlock, personalized music recommendations, spell-check in messages, navigation app traffic predictions, and school grading software. Ask: how many of these used AI? Most students are surprised by how deeply AI is already embedded in daily life.

### Explore (10-12 minutes)

Students form expert groups, each researching AI in a different domain: healthcare (diagnosis, drug discovery), transportation (autonomous vehicles, route optimization), education (adaptive learning, automated grading), entertainment (game AI, content generation), and agriculture (crop monitoring, yield prediction). Each group prepares a 3-minute briefing with concrete examples.

### Explain (10-12 minutes)

AI applications span virtually every industry. Healthcare AI analyzes medical images with superhuman accuracy, predicts disease outbreaks, and personalizes treatment plans. Transportation AI enables autonomous driving, optimizes logistics, and manages traffic flow. Education AI provides personalized tutoring, automates grading, and identifies at-risk students. Finance AI detects fraud, assesses credit risk, and algorithmically trades. These applications share common traits: they process vast data, identify patterns, and make predictions faster than humans.

### Elaborate (8-10 minutes)

Deep-dive case studies: how Google’s AlphaFold predicted protein structures (transforming biology), how AI-generated art challenges copyright law, and how AI surveillance raises privacy concerns. Examine the economic impact: job displacement versus job creation, new industries born from AI capabilities, and the growing demand for AI literacy. Discuss the difference between AI as a tool (augmentation) versus AI as a replacement (automation).

### Evaluate (5-8 minutes)

Students research one AI application in depth and write a balanced analysis covering: what the AI does, how it works at a high level, who benefits, what risks exist, and what regulations (if any) govern it. Presentation to the class with Q&A. Peer evaluation of analysis depth and balanced perspective.